

PARTICIPATION AGREEMENT

DISCLAIMER

This contract version constitutes solely an informal translation with no binding legal content whatsoever. In any cases of disputes regarding the rights and obligations of the parties under this contract and the interpretation of the contract the only German version is relevant.

Between

SPRIND GmbH

Lagerhofstr. 4, 04103 Leipzig, Germany

- *hereafter referred to as: "SPRIND" -*

and

(...)

- *hereafter referred to as: "Participant" -*

SPRIN-D

CONTENTS

Preamble

Part 1: Participation in the challenge

§ 1 Participation

§ 2 Timeline and Basis for the Challenge

§ 3 Compensation of the Participant

§ 4 Intellectual Property

Part 2: Objectives and Stages of the Challenge

§ 5 Goal of the Challenge

§ 6 Objectives and Reports regarding the Individual Stages

§ 7 Selection for Additional Stages

Part 3: Rights and Obligations during Participation

§ 8 Coaching

§ 9 Cooperation

§ 10 Withdrawal of the Participant during a Stage

§ 11 Conflicting Intellectual Property Rights and Applications for Intellectual Property Rights

Part 4: Miscellaneous Provisions

§ 12 Contract Termination and Contract Term

§ 13 Legal Liability

§ 14 Additional Assignments

§ 15 Disputes

§ 16 Evaluations

§ 17 Written form, legal venue, choice of law, contract language, severability clause

PREAMBLE

The economic and sustainable use of our resources will be crucial for production in the 21st century. To date, however, our production has been based almost entirely on using newly mined raw materials or resources rather than reusing existing products. In so doing, the extraction and processing of new raw materials and resources continues to be an enormous burden on the environment and society. Also, many supply chains remain dependent on raw materials and resources that could otherwise be harvested through local material cycles. Instead, a circular economy that manufactures new products with existing raw materials and resources facilitates more sustainable and resilient production.

More sustainable production would be possible by greatly reusing raw materials through biomanufacturing processes. For this to become reality, biomanufacturing processes must be developed to market maturity and integrated directly with other production processes. Recent scientific advances have led to new insights and methods that can significantly increase the performance of biomanufacturing processes and extrapolate new application possibilities. These advances are urgently needed because although alternative ways of producing a wide variety of products from the conventional petrochemical or chemical industry have been demonstrated in various development approaches, they have only succeeded in niche applications thus far. But the goal has to be to transition a large part of production to utilizing locally available (secondary) raw materials.

On the way to a circular economy based on biomanufacturing processes, there are numerous challenges to overcome. For example, since bioproduction methods are subject to different scaling laws than those in the conventional petrochemical or chemical industry, it is nearly only profitable to produce special, high-value products. To date, when it comes to simple biomanufactured products, current biomanufacturing processes are not competitive compared to centralized large-scale production in the chemical industry. An additional challenge is the limited range of microbes used in biomanufacturing. So far, companies have mainly relied on *E. coli* and *Saccharomyces cerevisiae*. While the genetic manipulation of microbes like these is easy, the production of certain biomanufactured products using these microbes is challenging. Many of the target products prove to be toxic to these microbes or cannot be produced in sufficient concentrations. In terms of process technology, *E. Coli* and *Saccharomyces cerevisiae* depend on sterile production sites and high-quality sugars as substrate. Production can generally also only be achieved in batches. Plus, investment efforts are aimed at making individual companies competitive with their niche products rather than developing new platform technologies. Production equipment, namely fermenters, is technologically outdated and is not optimized for biomanufacturing. Ultimately, it is currently not possible to feed microbes from decentralized feedstocks, such as waste streams, and then couple the products synthesis with modern manufacturing processes (e.g., additive manufacturing).

The Circular Biomanufacturing Challenge aims to develop solutions to these issues and essentially enable economic and sustainable production within a circular economy based on biotechnological processes.

The participant will have successfully demonstrated that it is able to participate in the Challenge. The basis of its participation is the following Participation Agreement.

PART 1: PARTICIPATION IN THE CHALLENGE

§ 1 PARTICIPATION

(1) The participant participates in a so-called pre-commercial competition (in the following: challenge) organized by SPRIND pursuant to this Participation Agreement. The goal of the challenge is to develop an

end-to-end prototype that processes various waste streams to provide a continuous feed-stock that allows microbes to produce a variety of material, which can feed into contemporary manufacturing processes such as additive manufacturing.

(2) The challenge was preceded by the search for a solution based on a tender dated 29th June 2023 and published on the homepage of SPRIND. This Participation Agreement is based on the tender documents. The participant has joined this search for a solution by submitting its solution outline attached as Exhibit A "_____". Based on the decision of the jury of the challenge dated _____ and attached as Exhibit B, this solution outline was selected for participation in the challenge. The participant undertakes to implement the solution outlined in Exhibit A in accordance with this Participation Agreement.

§ 2 TIMELINE AND BASIS FOR THE CHALLENGE

(1) The challenge shall take place in three stages:

- First stage
 - In the first stage, the competitors develop process engineering principles. The aim is to develop processes for the production of a purified biomanufactured product by microbial valorization of self-selected carbon-containing waste streams in order to lay the process engineering foundations for complete development.
 - The first stage begins on 01.11.2023 and ends on 31.10.2024.
 - The maximum contract amount for this stage is 1.500.000 € plus VAT.
 - At least four and at most eight competitors will participate in the first stage.
 - The requirement for an invitation to the first stage is the jury's assessment that the project proposed in the application has the potential to become a breakthrough innovation according to the selection criteria in Table 2: Selection criteria of the call. If more than four applicants meet this criterion, the jury will rank the applications according to the degree of fulfilment of the selection criteria.
- Second stage
 - In the second stage, practical implementation is to be driven forward on the basis of the techniques developed in the first year, an overall concept is to be developed and analyzed from a techno-economic perspective, and sub-components are to be further developed.
 - The second stage is expected to last from 01.11.2024 to 31.10.2025.
 - The maximum contract amount for this stage is expected to be 2.000.000 € plus VAT.
 - At least two and at most six competitors will participate in the second stage.

- The requirement for an invitation to the second stage is the jury's assessment that the project proposed in the application has the potential to become a breakthrough innovation according to the selection criteria in Table 2: Selection criteria of the call. If more than six applicants meet this criterion, the jury will rank the applications according to the degree of fulfilment of the selection criteria.
- Third stage
 - In the third stage, the competitors integrate the subcomponents into a functional end-to-end prototype.
 - The third stage is expected to last from 01.11.2025 to 31.10.2026.
 - The maximum contract amount for this stage is expected to be 2.500.000 € plus VAT.
 - At least two and at most two competitors will participate in the third stage.
 - The requirement for an invitation to the third stage is the jury's assessment that the project proposed in the application has the potential to become a breakthrough innovation according to the selection criteria in Table 2: Selection criteria of the call. If more than four applicants meet this criterion, the jury will rank the applications according to the degree of fulfilment of the selection criteria.

(2) With the jury decision (Exhibit B), the participant was invited to participate in the first stage. With the signature of this Agreement the participant agrees to participate in the challenge pursuant to this Agreement.

(3) The participant shall not acquire any claim to participate in additional stages (second and third stage) of the challenge on the basis of its participation in the first stage or on the basis of this Participation Agreement. Nor shall the participant acquire any claims to implementation of additional stages. Whether the execution of the second and third stage is appropriate or not shall be decided by the jury empaneled by SPRIND, among other things on the basis of the submitted competitive contributions. SPRIND may not implement the challenge as such or any additional stages in particular if

- the minimum number of participants is not reached;
- SPRIND does not have sufficient funds for budgetary or other financial reasons to implement the stages in a proper manner;
- the goal of the challenge is reached with an earlier result of a participant or the development has reached the point where any further implementation does no longer constitute research and development funding.

(4) If SPRIND decides to implement the respective additional stages, the participants selected by the jury will be requested to participate in the respective stages. In that case SPRIND and the participant will agree in writing on participation in the additional stages. In case the participant is selected to participate, the participant shall be free to participate in the additional stages. SPRIND shall be free to make changes to the provisions of this Agreement with regard to the additional stages of the challenge. If the participant is selected and agrees to participate in an additional stage, this Agreement shall apply to the additional stages subject to future changes. Negotiations between SPRIND and the participants may be part of the stages following stage 1. SPRIND will evaluate the submitted solutions from the respective additional stages according to the criteria set out in the procedure description, Table 2: Selection criteria. Before the start of the stages following stage 1, SPRIND may start negotiations with the aim of improving the solutions. Negotiations may cover the entire

content of the problem-solving approach, with the exception of any minimum requirements and evaluation criteria. Participants will then have the opportunity to adapt their solutions according to the outcome of the negotiations at all stages following stage 1.

§ 3 COMPENSATION OF THE PARTICIPANT

(1) The participant shall receive a compensation based on its offer - and/or based on its applications for the additional stages of the challenge - plus value added tax if applicable. This compensation shall be in lieu of all expenses necessary for the provision of the deliverables. No after-the-fact changes shall be allowed. The financial risk shall be borne by the participant.

(2) The participant shall be granted 50% of the compensation for the first stage at the start of the challenge. The remaining 50% for this stage shall be disbursed in two equal installments within six and nine months after the start of the challenge. The participant may indicate a different payment schedule for the first stage in its bid or its application for participation if such is justified for factual or other reasons.

(3) For the second and third stage, the compensation shall be disbursed to the participant in case of the participation in advance in four equal installments. The participants shall send the respective invoices to *buchhaltung@sprind.org*. The first installment shall be paid at the start of the stage, the additional installments in each case within three months. The participant may indicate a different payment schedule in its bid or its application for participation in the additional stages if such is justified for factual or other reasons.

(4) If the participant wishes to deviate materially from the calculated compensation indicated in its application, it must notify SPRIND accordingly. Corresponding invoices are to be sent by the participant to *buchhaltung@sprind.org*. SPRIND may deviate from the above payment plans in favor of the participants and arrange for payments to the participant earlier on the basis of credit notes.

(5) SPRIND reserves the right to specify price ceilings for the additional stages of the challenge.

(6) The participant undertakes to use all support granted by SPRIND on the basis of this challenge exclusively within the scope of this challenge (earmarking) and to provide proof of this to SPRIND upon request. In particular, it is not permitted to use such support services within the scope of this challenge that were granted by SPRIND or third parties on the basis of another challenge without the prior written consent of SPRIND.

§ 4 INTELLECTUAL PROPERTY

(1) The intellectual property developed during the challenge ("Results") shall be owned by the participant and SPRIND pursuant to the following dispositions:

1. The participant shall have the right to apply for intellectual property rights regarding the results of the challenge. The participant must inform SPRIND immediately if copy-rightable results are developed or if applications are submitted.
2. The participant shall have the right but not the obligation to publish the results in scientific journals. In case of publication, a proper reference to the cooperation with SPRIND must be included.

3. SPRIND shall receive a free, unrestricted, non-exclusive right to use all results. SPRIND shall also be entitled to commercially exploit, further develop or have research carried out on results found with third parties in accordance with No. 4 below.
4. The participant agrees towards SPRIND and by way of a genuine contract for the benefit of third parties to grant to any third party, non-exclusive licenses on a non-discriminatory basis for the commercial exploitation of the result(s) and for research and development on normal market terms. In doing so, the participant(s) will in general give priority to institutions located in the EU over institutions located outside the EU in order to take full account of the potential of the European Research Area. The foregoing obligation to grant licenses shall be suspended to the extent and for as long as SPRIND has consented in writing to exclusive licensing to one or more third parties. Such consent shall be subject to the condition that it is ensured that SPRIND and the participant do not violate EU state aid regulations by granting an exclusive license. No. 4 sentence 2 shall apply accordingly to the granting of an exclusive license.

(2) A result within the meaning of the above paragraph is any intellectual property even if it was developed mainly during the challenge and if the result is based overall on the results of the challenge. The portion of the work of the participant that took place after the challenge may be considered when determining the customary market conditions pursuant to the above Paragraph 1 No. 4.

(3) The participant must consider the above dispositions regarding the results in case of contracts with third parties. The participant undertakes in particular not to sign any contracts with third parties that violate the above regulations or limit, hinder or frustrate the implementation of the resulting rights of third parties or SPRIND. SPRIND shall have the right to notify third parties of the obligation of the participant to grant licenses pursuant to the above Paragraph 1 No. 4.

(4) The participant shall apply for copyrightable intellectual property rights at its own expense. If the participant decides against submitting an application, it must inform SPRIND accordingly, including the considerations that led to the decision. SPRIND may require that the participant apply for intellectual property rights unless the participant has a legitimate interest not to do so. This shall apply also if the application requires a third-party action within the meaning of the Employee Invention Act.

PART 2: OBJECTIVES AND STAGES OF THE CHALLENGE

§ 5 GOAL OF THE CHALLENGE

- (1) The goal of the challenge is to develop an end-to-end prototype that processes various waste streams to provide a continuous feedstock that allows microbes to produce a variety of material, which can feed into contemporary manufacturing processes such as additive manufacturing
- (2) The production plant to be developed should seamlessly integrate the complete value chain from the processing of carbon-containing waste into a substrate, through the microbial

extraction of biomanufactured products, to the production of commercializable end products.

§ 6 OBJECTIVES AND REPORTS REGARDING THE INDIVIDUAL STAGES

(1) The following objectives are established for the stages:

- At the end of Stage 1, teams are expected to be able to process three self-selected carbonaceous waste streams into a substrate that allows for the continuous feeding and growth of an unconventional microbial strain (non-E. Coli and *Saccharomyces cerevisiae* strains). The process must be designed to be as sustainable, efficient, and cost-effective as possible. The yield coefficient (amount of synthesis product per substrate consumed) shall reach a value of at least 0.5 g/g. The synthesis product shall reach a minimum concentration of 10g/L relative to the fermentation medium.
- At the end of Stage 2, teams are expected to submit a competitive design (including cradle-to-grave, techno-economic and landscape analysis) for a continuous biomanufacturing process. The design should cover the complete value chain from the carbon-containing waste stream to the commercializable end product.
- At the end of Stage 3, teams are expected to have developed a functional, scalable, and demonstration-ready prototype biomanufacturing facility for manufacturing of three unique, commercializable products from carbon-containing waste streams.

(2) A report regarding the completed stage must be submitted at the end of each stage. The report must include

- an explanation as to whether the objective of the respective stage described in the application has been achieved,
- a description of all the work performed and a summary description of the results,
- a list of the intellectual property developed in the process (know-how, data, inventions etc.)
- if applicable, a list of publications.

Furthermore, the following shall apply:

- The first stage report should include experimental methodology and data as well as conceptual considerations related to the objectives stated in §6. The report should identify which content points are missing for the development of a prototype in accordance with the stage 3 goal and present a plan on how these are to be fulfilled in the next stage.
- The report for the second stage should describe the complete design of the finished end-to-end prototype and the individual subcomponents. This must include a description of the biotechnological processes, the input and output processes including the intermediate products and the efficiency of the production system (materials, energy). In addition, the report must contain a complete techno-economic analysis, including the supply chains for raw materials and the personnel and training requirements.
- The final report must describe how the construction of the complete, operational, and functional prototype is scalable.

(3) The structure and content of the report for the respective stage shall be due irrespective of whether the participant decides to apply for participation in additional stages.

(4) The report must be submitted in digital form to SPRIND one month before the end of the respective stage. SPRIND will provide the participant with other requirements in terms of form or content and with templates for the report in a timely fashion.

§ 7 SELECTION FOR ADDITIONAL STAGES

(1) If the participant decides to apply for participation in the additional stages of the challenge, the report for the preceding stage must be accompanied by the following:

- a detailed work schedule and indication of a project-specific objective for the next stage,
- a declaration as to whether and to what extent the team composition has changed or will change,
- a price bid for the next stage.

(2) The selection will be made on the basis of the selection criteria and procedures specified in the tender documents of the challenge. SPRIND will advise the participants of the detailed conditions regarding the application for additional stages in a timely fashion.

PART 3: RIGHTS AND OBLIGATIONS DURING PARTICIPATION

§ 8 COACHING

(1) The coach will assist SPRIND by providing help to the participants and the jury in their decision-making. The coach will provide the jury with a written assessment regarding the participant at each stage of the challenge, including information about the development of the proposed solution, its economic viability and team competence.

(2) The coach advises the participant regarding the organization and performance of experiments. The coach supports the participant in negotiating subcontracts, investments and in setting up cooperation agreements. If possible and desirable, the coach provides assistance with any spin-off from the academic environment, with patenting strategy, with liquidity management and with the recruitment of suitable employees.

(3) The participant is obliged to work with the coach, but the coach shall not have any command authority over the participant. The participant shall not have any claims to the provision of coaching services..

§ 9 COOPERATION

(1) SPRIND shall have the right to observe the progress of the work and inspect all documents necessary for this purpose, including notes about materials and work performed, and to monitor compliance with the technical specifications.

(2) If SPRIND provides suggestions, recommendations and other contributions (e.g. inventions) to the participant in order to further participation in the challenge, the participant must take these into consideration, if possible.

§ 10 WITHDRAWAL OF THE PARTICIPANT DURING A STAGE

If the participant does not complete a stage of the challenge or does not or no longer participate in the stage even though it issued the respective declaration, the participant must inform SPRIND immediately. A stage is considered as not completed if the report due under § 6 has not been submitted at all, only in part or not on time.

§ 11 CONFLICTING INTELLECTUAL PROPERTY RIGHTS AND APPLICATIONS FOR INTELLECTUAL PROPERTY RIGHTS

The participant must immediately notify SPRIND of any intellectual property rights or applications for intellectual property rights that are in conflict with the exploitation of the research and development results, unless they are already mentioned in the tender documents. In conflict are such intellectual property rights or applications for intellectual property rights of third parties as are required for any exploitation of the research and development results. The participant must indicate the conditions under which, in its opinion, such use is still expected to be possible. This obligation shall only apply to the extent that the participant has become aware of the conflicting property rights and property right application.

PART 4: MISCELLANEOUS PROVISIONS

§ 12 CONTRACT TERMINATION AND CONTRACT TERM

(1) The contract term of this Participation Agreement begins on 01.11.2023 and ends on 31.10.2024, unless the Parties agree to its application in additional stages. The termination of the Participation Agreement for any reason shall not affect the regulation in § 4 of this Participation Agreement. In addition, the participant shall immediately surrender all results, records, data and information about its research work in the respective stage..

(2) SPRIND may terminate the Agreement without notice for important reasons. An important reason is present in particular if

- it turns out that the participant gave incorrect information or if the conditions or requirements specified in the tender are not or no longer present or cannot or no longer be satisfied;
- the opening of bankruptcy proceedings regarding the assets of the participant or one of its subcontractors has been requested and this request is not withdrawn within six weeks, or if a provisional bankruptcy trustee has been appointed or if the opening of the bankruptcy proceedings was ordered but dismissed for lack of assets;
- in the events of § 10 of this Participation Agreement;
- the participant fails to issue a notice pursuant to § 10 but still withdraws from the challenge;
- the participant fails to issue a notice pursuant to § 3 Para. 4 or if the deviations planned by the participant are material enough to fear that the objective of the respective stage cannot be achieved;
- the participant does not fulfill its obligation for the issuance of reports;

- the participant seriously or repeatedly violated any of the obligations specified in this Agreement or its Exhibits or does not provide the services indicated in the respective applications without the consent of SPRIND, or materially changes them without the consent of the SPRIND;
- violations of safety regulations, such as the Genetic Engineering Act or the Biological Substances Ordinance or the corresponding national regulations, occur during the provision of services.;
- the participant accepts subsidies from the Federal Republic of Germany in addition to SPRIND for the same project and the work processes indicated in its bid/applications, in particular from the Federal Ministry of Education and Research.
- the participant does not use the funds and support granted by SPRIND in accordance with their earmarking or, conversely, uses funds and support granted by SPRIND on the basis of other challenges and projects in the context of this challenge.

(3) If SPRIND terminates the contract for important reasons, the participant shall repay the funds already received from SPRIND for the respective level. Insofar as the participant can prove that the services provided by it in the respective stage up to the time of the extraordinary termination are not worthless for SPRIND, it shall be entitled to the remuneration attributable to the non-worthless part, insofar as this does not exceed any remuneration customary in the market. In particular, these services are not worthless if they are used by other participants in the challenge.

§ 13 LEGAL LIABILITY

SPRIND shall not be legally liable for any losses incurred by the participant or third parties which arise from their participation in the challenge. The participant must hold SPRIND harmless if SPRIND is held liable for such losses. Sentence 1 shall not apply if SPRIND caused such losses willfully.

§ 14 ADDITIONAL ASSIGNMENTS

When awarding additional development assignments, SPRIND will invite the participant to participate in the challenge. However, under this Participation Agreement participation in the challenge does not create a legal claim of the participant for being granted such assignments, nor will the participant, in case of being awarded an additional assignment, enjoy any preferential treatment when providing the finished product or finished service in commercial volumes to a public client in the respective Member State.

§ 15 DISPUTES

(1) In case of disputes regarding technical or scientific issues or regarding questions as to whether and to what extent the material conditions for being granted a user or beneficial use right are present, two arbitrators shall be appointed, of which SPRIND and the participant shall nominate one each. If no agreement is reached between the two arbitrators, they shall select jointly a third party as chairman. If the arbitrators do not reach an agreement within one month after an arbitrator nominated a person as chairman for the first time, the President of the ICC Leipzig shall designate a chairman. The arbitrator committee then shall make a decision with a two-thirds majority; if no majority emerges, the chairman shall cast the

deciding vote. The provisions of the Code of Civil Procedure shall apply to the settlement of the costs of the arbitration accordingly.

(2) In all other disputes an amicable settlement shall be sought before involving a court of law. The right of the Parties to enter into arbitration agreements shall not be abridged.

§ 16 EVALUATIONS

The participant shall participate, as needed, in evaluations conducted by SPRIND itself or commissioned or approved by SPRIND.

§ 17 WRITTEN FORM, LEGAL VENUE, CHOICE OF LAW, CONTRACT LANGUAGE, SEVERABILITY CLAUSE

(1) Changes to this Agreement and its Exhibits shall require the written form.

(2) If admissible under the law, Leipzig shall be the legal venue. This Agreement shall be governed by the laws of the Federal Republic of Germany without the possibility of recourse to the UN Convention on the International Sale of Goods. The accompanying English version of this Agreement is only for information purposes and carries no legal obligation.

(3) In the event that an individual provision of this Agreement is invalid or unenforceable, such provision shall not affect the validity of the Agreement as a whole. Such invalid or unenforceable provision shall be replaced by a valid and enforceable provision whose economic effect is closest to the effect of the invalid or unenforceable provision. The above shall also apply if this Agreement turns out to contain omissions.